

Quadratic Simultaneous Equations

AQA Higher · GCSE Grade 8–9

Exam Practice Booklet (Dyslexia-adapted)

Name: _____ Class: _____ Date: _____

Marks available: 33 My score: _____ / 33

What you need to be able to do

Solve a pair of simultaneous equations where one is linear (a straight line) and one is quadratic (a curve), using substitution. Each pair has two answer points — find both, and give each as an (x, y) pair.

Method steps — keep this in view

Method steps — refer back to this while you work:

1. Step 1: Make y the subject of the straight-line (linear) equation.
2. Step 2: Substitute that into the curve (quadratic) equation.
3. Step 3: Move everything to one side so the equation equals 0.
4. Step 4: Solve the quadratic — factorise, or use the formula.
5. Step 5: Put each x back in to find its y. Write your answers in pairs and check them.

Worked example

Solve: $y = x^2 - 2x + 3$ and $y = x + 3$

Step 1-2: The line already gives y , so substitute: $x^2 - 2x + 3 = x + 3$

Step 3: Move to one side: $x^2 - 3x = 0$

Step 4: Factorise: $x(x - 3) = 0$ $x = 0$ or $x = 3$

Step 5: Find y from $y = x + 3$: when $x = 0$, $y = 3$; when $x = 3$, $y = 6$

Answers: $(0, 3)$ and $(3, 6)$

Question 1

[3 marks]

Solve the simultaneous equations.

$$y = x^2 \text{ and } y = x + 12$$

Question 2**[3 marks]**

Solve the simultaneous equations.

$$y = x^2 + 2x \text{ and } y = 3x + 6$$

Question 3**[3 marks]**

Solve the simultaneous equations.

$$y = x^2 - 5x + 6 \text{ and } y = x - 2$$

Question 4**[4 marks]**

Solve the simultaneous equations.

$$x^2 + y^2 = 25 \text{ and } y = x - 1$$

Question 5**[4 marks]**

Solve the simultaneous equations.

$$x^2 + y^2 = 10 \text{ and } x + y = 4$$

Question 6**[4 marks]**

Solve the simultaneous equations.

$$y = 2x^2 + 5x - 3 \text{ and } y = x - 3$$

Question 7**[5 marks]**

Solve the simultaneous equations. Give your answers in surd form (leave in your answer).

$$x^2 + y^2 = 9 \text{ and } y = x + 1$$

Question 8**[5 marks]**

A rectangle has an area of 12 cm^2 and a perimeter of 14 cm.

By forming and solving a pair of simultaneous equations, find the length and width of the rectangle.

Answers (teacher copy)

Q1: (4, 16) and (-3, 9)

Q2: (3, 15) and (-2, 0)

Q3: (2, 0) and (4, 2)

Q4: (4, 3) and (-3, -4)

Q5: (1, 3) and (3, 1)

Q6: (0, -3) and (-2, -5)

Q7: $x = \frac{-1 + \sqrt{17}}{2}$, $y = \frac{1 + \sqrt{17}}{2}$ and $x = \frac{-1 - \sqrt{17}}{2}$, $y = \frac{1 - \sqrt{17}}{2}$

Q8: length 4 cm, width 3 cm (from $lw = 12$ and $l + w = 7$)

Dyslexia adaptations applied to this booklet

- OpenDyslexic typeface (Comic Sans MS fallback)
- Base font-size 16px with proportional section sizing
- Line-height 1.7 (reducedDensity)
- Letter-spacing 0.04em, word-spacing 0.16em
- Ragged-right (left-aligned) text, never justified
- Cream #FFF8E7 colour overlay (COBS / BDA-aligned, reduces contrast glare)
- Method-steps box pinned before the first question (working-memory reference)
- Answer space expanded 1.5x (extraAnswerLinesMultiplier)
- Maximum 5 questions per page (page breaks added)

Original questions authored for this booklet on the same AQA topic and at the same Grade 8–9 difficulty as the source on [yesgenie.com](https://www.yesgenie.com). Not a copy of the source booklet's questions.